

Basic Virology

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1. History of Virology

1.Ancient Awareness of Viruses: Evidence suggests that ancient peoples recognized the effects of viral infections, with written records dating back to around 3700 BC in Egypt, depicting symptoms of paralytic poliomyelitis.



Figure 1 :Perhaps the first written record of a virus infection consists of a hieroglyph from Memphis, the capital of ancient Egypt, drawn in approximately 3700BC.

2. Notable Historical Figures:

1. **Pharaoh Siptah** (1200-1193 BC): His mummy shows signs consistent with paralytic poliomyelitis.
2. **Pharaoh Ramses V** (died 1196 BC): Believed to have died from smallpox, as indicated by pustular lesions on his mummy.



Figure 2: The Pharaoh Siptah ruled Egypt from 1200-1193 BC

3. **Smallpox and Variolation:** By 1000 BC, smallpox was endemic in China, leading to the development of variolation, a practice involving the inhalation or inoculation of materials from smallpox lesions to confer immunity.
4. **Edward Jenner's Vaccination:** On May 14, 1796, Jenner successfully vaccinated James Phipps using cowpox material. This marked a significant advancement in preventive medicine.
5. **Germ Theory of Disease:** In the 1880s, Robert Koch and Louis Pasteur established the germ theory, highlighting the role of microorganisms in disease. Koch's postulates provided a framework for linking specific pathogens to specific diseases.

➤ **Koch's postulates: Definition of a pathogen**

- i. The agent must be present in every case of the disease.**
- ii. The agent must be isolated from the host & grown in vitro.**
- iii. The disease must be reproduced when a pure culture of the agent is inoculated into a healthy susceptible host.**
- iv. The same agent must be recovered once again from the experimentally infected host.**

Bacteriophage

Frederick Twort (1877-1950), and Felix d'Herelle (1873-1949) were the first to recognize viruses which infect bacteria, which called bacteriophages (eaters of bacteria). Luria, Delbruck & many others utilized these viruses as model systems to investigate many aspects of virology, including virus structure, genetics, replication, etc.

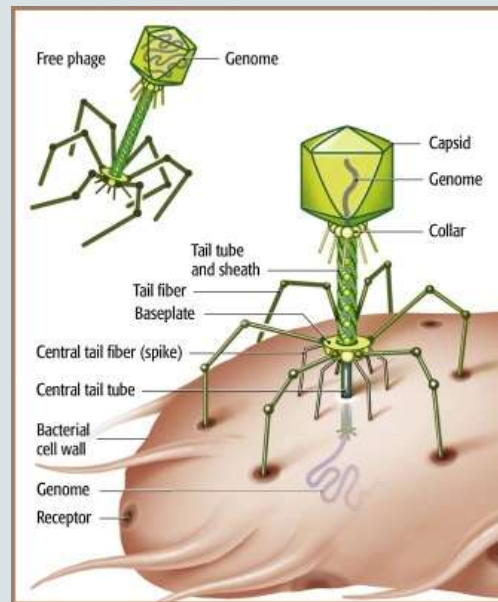


Figure 3 : Structure of Bacteriophage

2. Definition of a Virus

•**Definition:** Infectious entities with an infectious phase, characterized by:

- i. Possessing only one type of nucleic acid (RNA or DNA).
- ii. Multiplying through their genetic material.
- iii. Unable to undergo binary fission.

•**Nature:** Acellular organisms that are intracellular parasites, dependent on host cells for replication.

•**Replication:** Use host cellular machinery to synthesize components that assemble into virions, which protect the viral genome and facilitate transmission to other cells.

Viroids

- Definition:** Small, circular RNA molecules (200-400 nucleotides) without a capsid or envelope.
- Characteristics:** Associated with certain plant diseases and replicate as obligate intracellular parasites, similar to viruses.

Prions

- Definition:** Infectious agents composed solely of protein, lacking nucleic acid.
- Confusion:** The prion protein and its encoding gene are present in normal cells.
- Associated Diseases:** Linked to neurodegenerative diseases like Creutzfeldt-Jakob disease, scrapie in sheep, and bovine spongiform encephalopathy (BSE) in cattle.

3. Basic characteristics of Viruses

- **Structure:** Viruses are heterogeneous but share a basic structure comprising protein and nucleic acid (either DNA or RNA).
- **Replication:** Unlike binary fission, viruses replicate through a burst of virus particles. They assemble from pre-formed components rather than growing through division.
- **Size:** Viruses are filterable agents, typically ranging from 20 to 200 nm, which is smaller than the resolving power of light microscopes.
- **Nature:** Obligate intracellular parasites that rely entirely on host cell machinery for replication and energy.
- **Genome:**
 - Contains either DNA or RNA, but not both.
 - Most DNA viruses are double-stranded, while most RNA viruses are single-stranded (ssRNA) with variations in sense:
 - **Positive-sense ssRNA:** Can be directly used as mRNA for protein synthesis.
 - **Negative-sense ssRNA:** Must be converted into mRNA before protein synthesis.
- **Virion:** Virus particles (virions) do not grow or divide; they are assembled from synthesized components.
- **Inert Nature:** Viruses lack the genetic information necessary for metabolic processes or protein synthesis, making them non-living entities.

Comparison with Bacteria

- Metabolism:** Bacteria are larger and capable of carrying out their own metabolic processes and coding for their own enzymes.
- Enzymatic Differences:** Bacterial enzymes differ from eukaryotic enzymes, allowing for targeted antibiotic treatment.
- Intracellular Lifestyle:** Some bacteria (e.g., mycoplasma, rickettsia, chlamydia) can also enter eukaryotic cells and become parasites, but they provide their own enzymes necessary for replication.
- Control Mechanisms:** Bacterial control mechanisms are generally easier to develop compared to those for viruses due to their metabolic independence.

Control measures for microorganisms include capitalizing on our knowledge of:						
	Growth on artificial media	Division by binary fission	Whether they have both DNA and RNA	Whether they have ribosomes	Whether they have muramic acid	Their sensitivity to antibiotics
Bacteria	Yes	Yes	Yes	Yes	Yes	Yes
Mycoplasma	Yes	Yes	Yes	Yes	No	Yes
Rickettsia	No	Yes	Yes	Yes	Yes	Yes
Chlamydia	No	Yes	Yes	Yes	No	Yes
Viruses	No	No	No	No *	No	No
* The arenavirus family (an RNA virus family) appears to package ribosomes 'accidentally'. The packaged ribosomes appear to play no role in viral protein synthesis.						

Table 1 : Differences among different microorganisms

4. Consequences of Viral Properties

1. Non-Living Status:

- Viruses challenge the traditional definitions of life because they do not exhibit key characteristics associated with living organisms:
 1. **No Respiration:** Viruses do not metabolize energy.
 2. **Lack of Irritability:** They do not respond to stimuli.
 3. **No Movement:** Viruses do not have the ability to move independently.
 4. **No Growth:** They do not grow or develop; instead, they assemble from components.
 5. **Reproduction:** While they reproduce, they rely on host cells to do so and can adapt to new hosts.

2. Infectious Nature:

- For viruses to persist in nature, they must be infectious, meaning they must be capable of spreading from one host to another.

3. Utilization of Host Cell Processes:

- Viruses must hijack host cellular processes to produce their components, including:
 1. Viral messenger RNA (mRNA).
 2. Viral proteins.
 3. Copies of their genomes.

4. Encoding Required Processes:

- Viruses need to encode any processes that are not provided by the host cell to facilitate their replication and assembly.

5. Self-Assembly of Components:

- Viral components must be capable of self-assembling into complete virions without further assistance.

5. Shape and Size

- The size and shape of viruses are primarily determined by the amount and arrangement of their proteins and nucleic acids.

1.Nucleoprotein/Nucleocapsid:

1. The proteins and nucleic acids of each virus class assemble into a structure called a nucleoprotein or nucleocapsid.
 2. Some viruses have multiple layers of protein surrounding the nucleic acid, while others possess a lipoprotein membrane (envelope) derived from the host cell's membrane.
- The specific properties of the protein and nucleic acid constituents are unique to each class of virus, influencing their overall structure.

2.Size Variability:

1. Viruses vary in diameter from 20 nanometers (nm) to 250-400 nm.
2. Only the largest and most complex viruses can be observed under a light microscope, even at the highest resolution.

➤ **Shape Considerations:**

- When determining the size of a virus, its shape must also be considered, as different virus classes exhibit distinctive shapes.

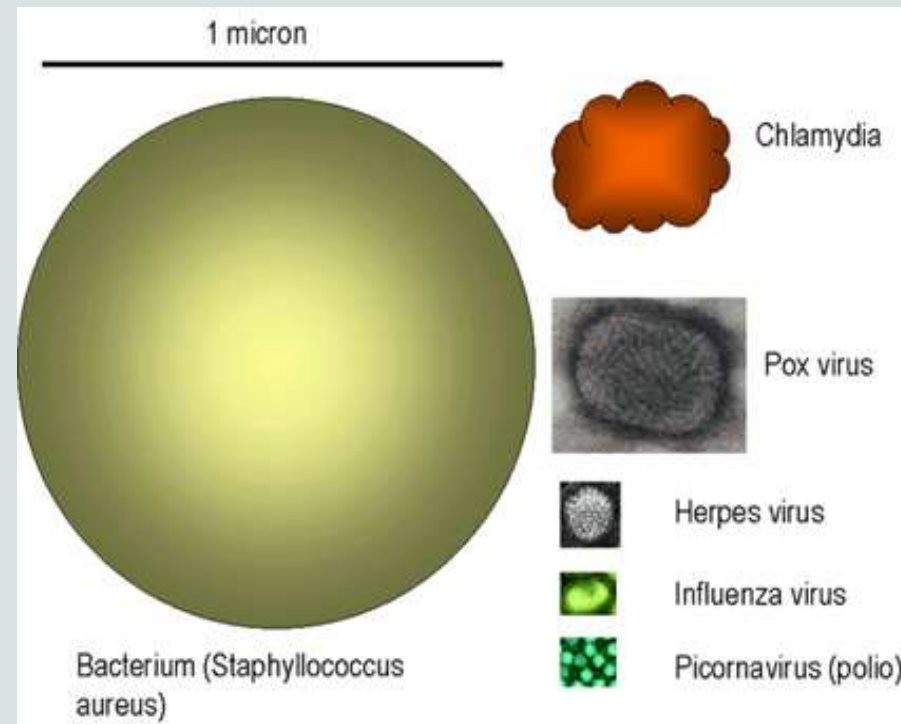


Figure 4: different shapes and sizes of viruses

➤ Shapes of Viruses

1. Predominant Shapes:

1. Rod-Shaped (Filamentous):

1. Characterized by a linear arrangement of nucleic acid and protein subunits.
2. Commonly found in many plant viruses and some bacterial viruses.

2. Spherical (Icosahedral):

1. Typically appear as 20-sided polygons (icosahedra).
2. Many viruses exhibit this geometric shape, especially among animal viruses.

2. Complexity in Bacteriophages:

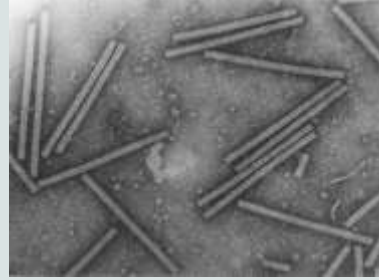
1. Larger and more complex bacteriophages often contain double-stranded DNA.
2. These viruses may exhibit a combination of both filamentous and polygonal shapes.

3. Diversity of Shapes:

1. Viruses can adopt various structures beyond the basic rod and spherical forms, reflecting their genetic makeup and evolutionary adaptations.



Spherical



Rod-shaped



Brick-shaped



Tadpole-shaped



Bullet-shaped



Filament

6. Virus structure

1. Size:

- Viruses can be as small as 20 nanometers in diameter (e.g., Parvoviridae) and can extend to several hundred nanometers in length (e.g., Filoviridae).

2. Structure:

- Nucleic Acid Genome:** Viruses possess either RNA or DNA as their genetic material.
- Capsid:** This is the protective protein coat that encases the nucleic acid.
- Nucleocapsid:** The combination of the nucleic acid genome and the capsid.
- Symmetry:** The nucleocapsid can exhibit different types of symmetry:
 - Icosahedral:** A spherical shape with 20 triangular faces.
 - Helical:** A spiral shape.
 - Complex:** A combination of different shapes.

3. Envelope:

- Some viruses have an outer lipid envelope, while others do not. The presence of an envelope can affect the virus's ability to infect host cells and its stability in the environment.

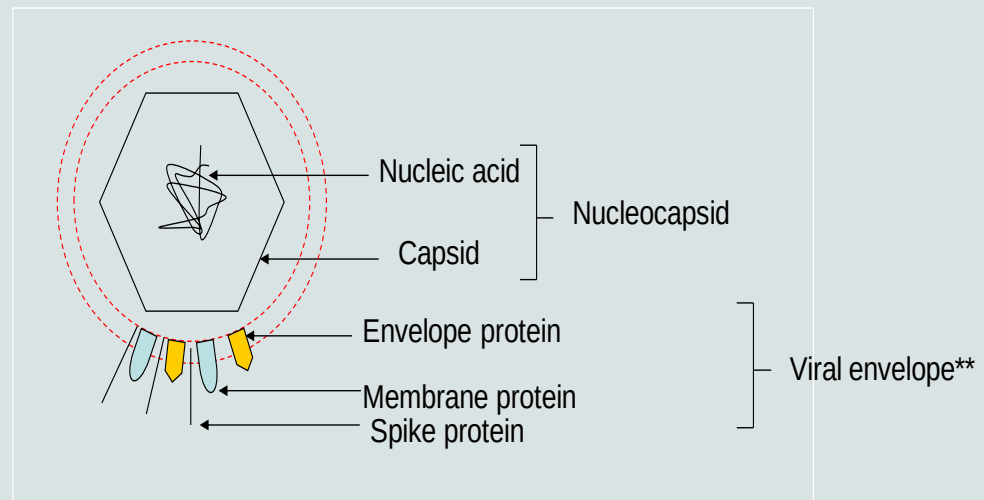


Figure 5 : Schematic overview of the structure of viruses

** does not exist in all viruses

- Enveloped viruses have a unique mechanism for acquiring their outer lipid envelope, which significantly influences their infectivity and interaction with host cells

1.Acquisition of Envelope:

1. Enveloped viruses obtain their envelope by budding from host cell membranes. This can occur through:
 - 1. Plasma Membrane:** The virus can bud directly through the outer membrane of the host cell.
 - 2. Internal Membranes:** In some cases, the envelope may originate from internal structures, such as the Golgi apparatus or the nuclear membrane.

2.Release Mechanism:

1. Unlike some viruses that destroy the host cell upon release, enveloped viruses can exit the cell without necessarily killing it. This budding process allows for:
 - 1. Persistent Infections:** The ability to leave the cell without causing death means that some enveloped viruses can establish long-term infections within the host.

3. Infectivity:

- The infectivity of enveloped viruses is highly dependent on the integrity of their envelope. The viral attachment proteins, which are crucial for recognizing and binding to host cell receptors, are located in the envelope.
- **Impact of Damage:** Agents like alcohols and detergents can damage the viral envelope, leading to reduced infectivity. This is why such agents are often used in disinfectants to control viral spread.

Structure

i. Viral Core

- The viral nucleic acid genome, in the center of the virion: Control the viral heredity and variation, responsible for the infectivity

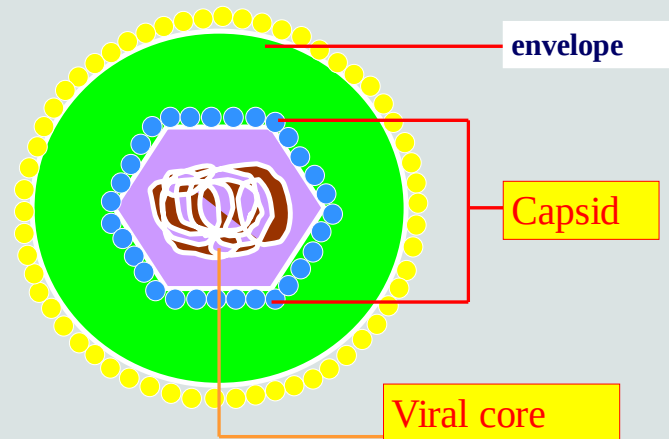


Figure 6: viral core

The viral nucleic acid genome plays a crucial role in the biology of viruses. **Location:**

1. The viral nucleic acid genome is typically found at the center of the virion, surrounded by the capsid or nucleocapsid.

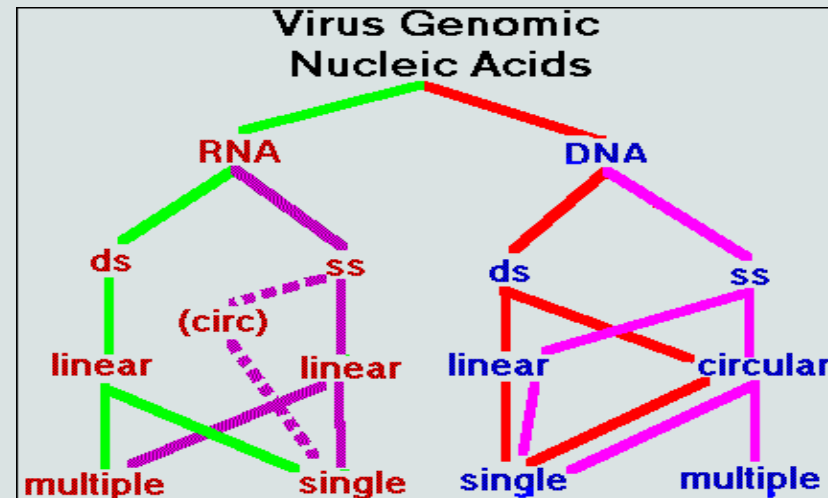
1.Functions:

1. **Control of Viral Heredity:** The genome contains the genetic information necessary for the replication and propagation of the virus. It dictates the traits passed on to progeny viruses during replication.
2. **Variation:** The nucleic acid can undergo mutations, leading to genetic variation. This variation can result in different viral strains, influencing their behavior, infectivity, and ability to evade the host's immune response.
3. **Responsibility for Infectivity:** The genome encodes proteins essential for the virus's ability to infect host cells, including:
 1. **Viral Attachment Proteins:** These proteins facilitate the binding of the virus to host cell receptors.
 2. **Replication Proteins:** Enzymes that help replicate the viral genome once inside the host cell.

2.Types of Nucleic Acids:

1. Viral genomes can be composed of either DNA or RNA, and they can exist in various forms, including single-stranded or double-stranded, linear or circular.

- ❖ **Genome**: The genome of a virus can be either DNA or RNA;
- ✓ DNA-double stranded (ds): linear or circular
- ✓ Single stranded (ss) : linear or circular
- ✓ RNA- ss: segmented or non-segmented
- ✓ RNA- ds: linear (only reovirus family)



ii. Viral Capsid

- The protein shell, or coat, that encloses the nucleic acid genome. It is functions:
 - a. Protect the viral nucleic acid.
 - b. Participate in the viral infection.
 - c. Share the antigenicity

iii. Nucleocapsid

- The cores of a virus particle consisting of the genome plus a complex of proteins, these complex proteins are structural proteins + non- structural proteins (Enzymes &Nucleic acid binding proteins).

iv. **Symmetry of Nucleocapsid**

1. Cubic (icosahedral):

Have 20 faces, each an equilateral triangle. Eg. Adenovirus

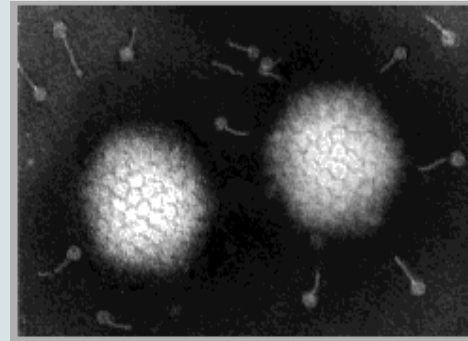
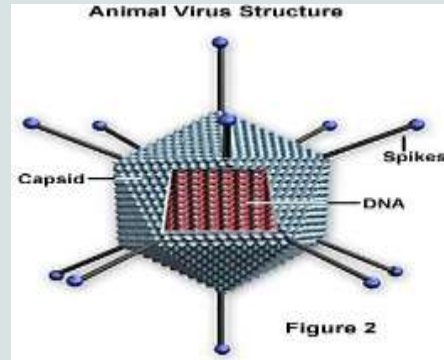


Figure 7: Cubic (icosahedral)

2. Helical

- Protein binds around DNA/RNA in a helical fashion e.g. Coronavirus
- Protein subunits can interact with each other and with the nucleic acid to form a coiled, ribbon-like structure.
- The best studied virus with helical symmetry is the non-enveloped plant virus, tobacco mosaic virus

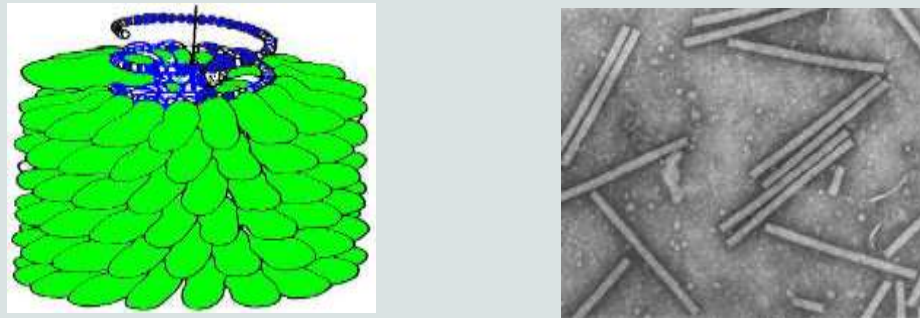


Figure 8: Helical

3.Complex

- Is neither cubic nor helical, these are regular structures, but the nature of the symmetry is not fully understood. Examples include the poxviruses

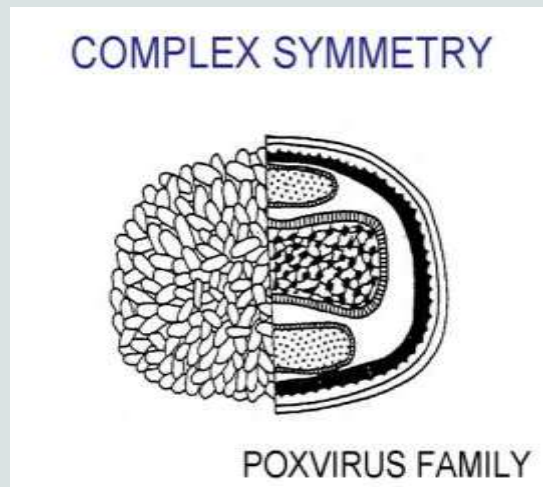


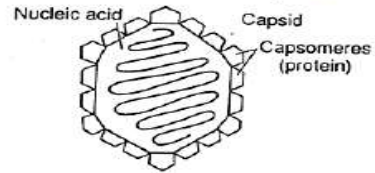
Figure 9: Complex

7. Five basic structural forms of viruses in nature

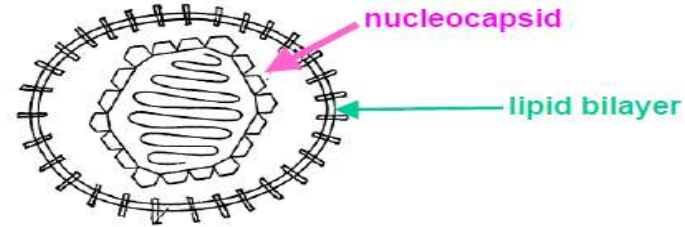
1. **Naked icosahedral** e.g. poliovirus, adenovirus, hepatitis A virus
2. **Naked helical** e.g. tobacco mosaic virus. So far no human viruses with this structure are known.
3. **Enveloped icosahedral** e.g. herpes virus, yellow fever virus, rubella virus
4. **Enveloped helical** e.g. rabies virus, influenza virus, parainfluenza virus, mumps virus, measles virus.
5. **Complex** e.g. poxvirus

5 BASIC TYPES OF VIRAL STRUCTURE

icosahedral nucleocapsid

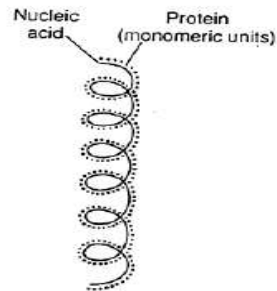


ICOSAHEDRAL

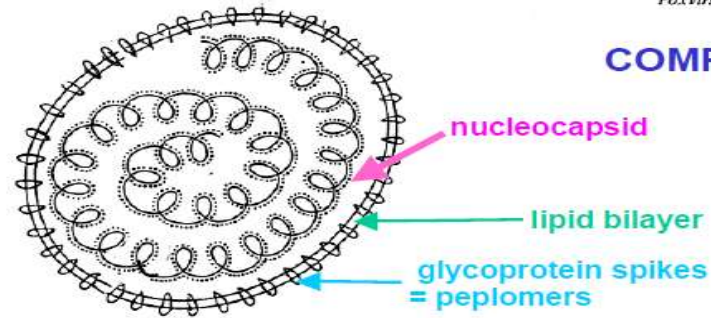


ENVELOPED ICOSAHEDRAL

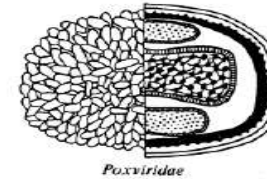
helical nucleocapsid



HELICAL



ENVELOPED HELICAL



COMPLEX

Figure 10: Five basic structural forms of viruses in nature

Naked viruses

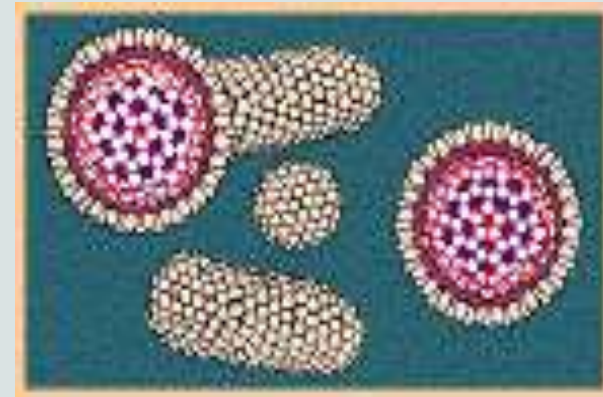
➤ Properties of naked viruses

- Stable in hostile environment.
- Not damaged by drying, acid, detergent, and heat.
- Can survive in the gastrointestinal tract.
- Retain infectivity on drying.
- Survive well on environmental surfaces.
- Must kill host cells for release of mature virus particles
- Can spread easily via hands, dust, fomites, etc.
- Can stay dry and still retain infectivity.
- Humoral antibody response may be sufficient to neutralize infection

■ Envelope viruses

Envelope is a lipid-containing membrane that surrounds some viral particles. It is acquired during viral maturation by a budding process through a cellular membrane; Viruses-encoded glycoproteins are exposed on the surface of the envelope. It is

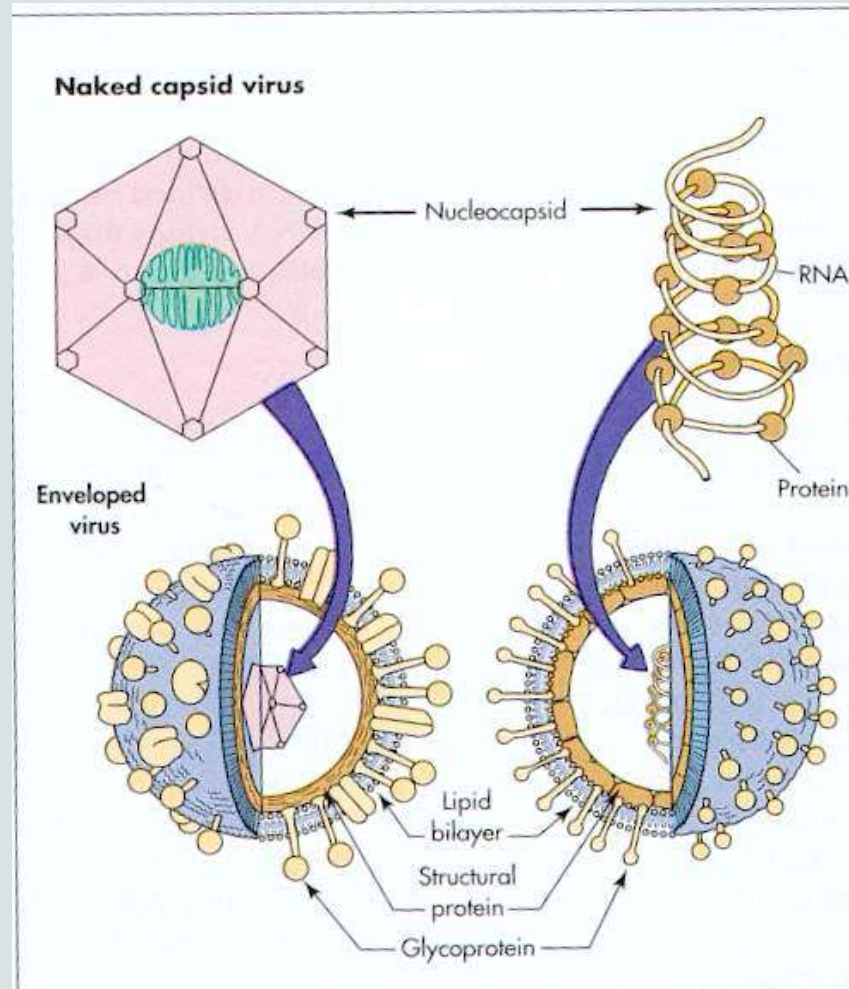
function for Antigenicity, Infectivity, and Resistance



Envelope viruses

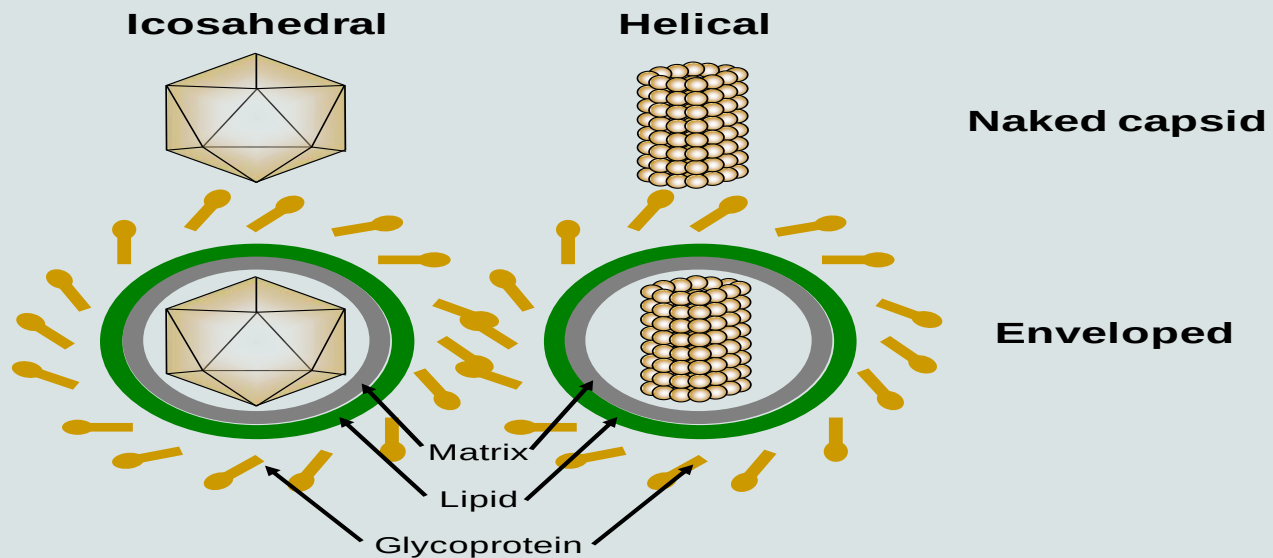
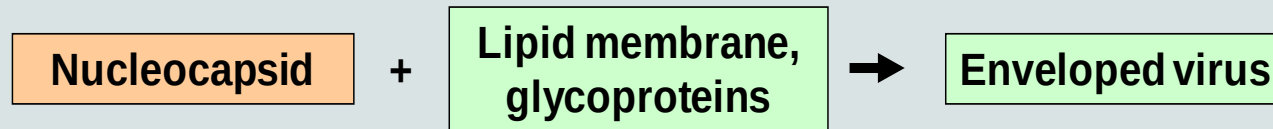
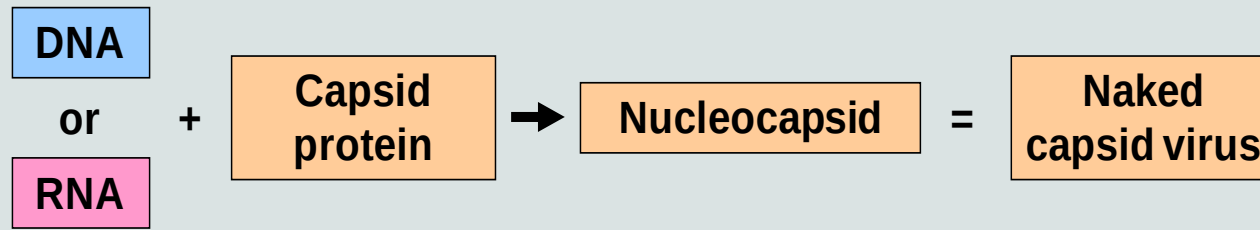
➤ **Properties of envelope viruses**

- Unstable in dry environment.
- Damaged by drying, acid, detergent, and heat.
- Must stay wet during transmission.
- Transmission in large droplets and secretions.
- Pick up new cell membrane during multiplication.
- Insert new virus-specific proteins after assembly.
- Virus is released by budding.
- Cannot survive in the gastrointestinal tract
- Do not need to kill cells in order to spread
- May require both a humoral and a cellular immune response



**Naked
Virus**

**Enveloped
Virus**



7. Classification of virus

- The internationally agreed system is based on structure/composition of the virus particles (virion), in some cases, the mode of replication is also important in classification. Viruses are classified into various families on this basis.

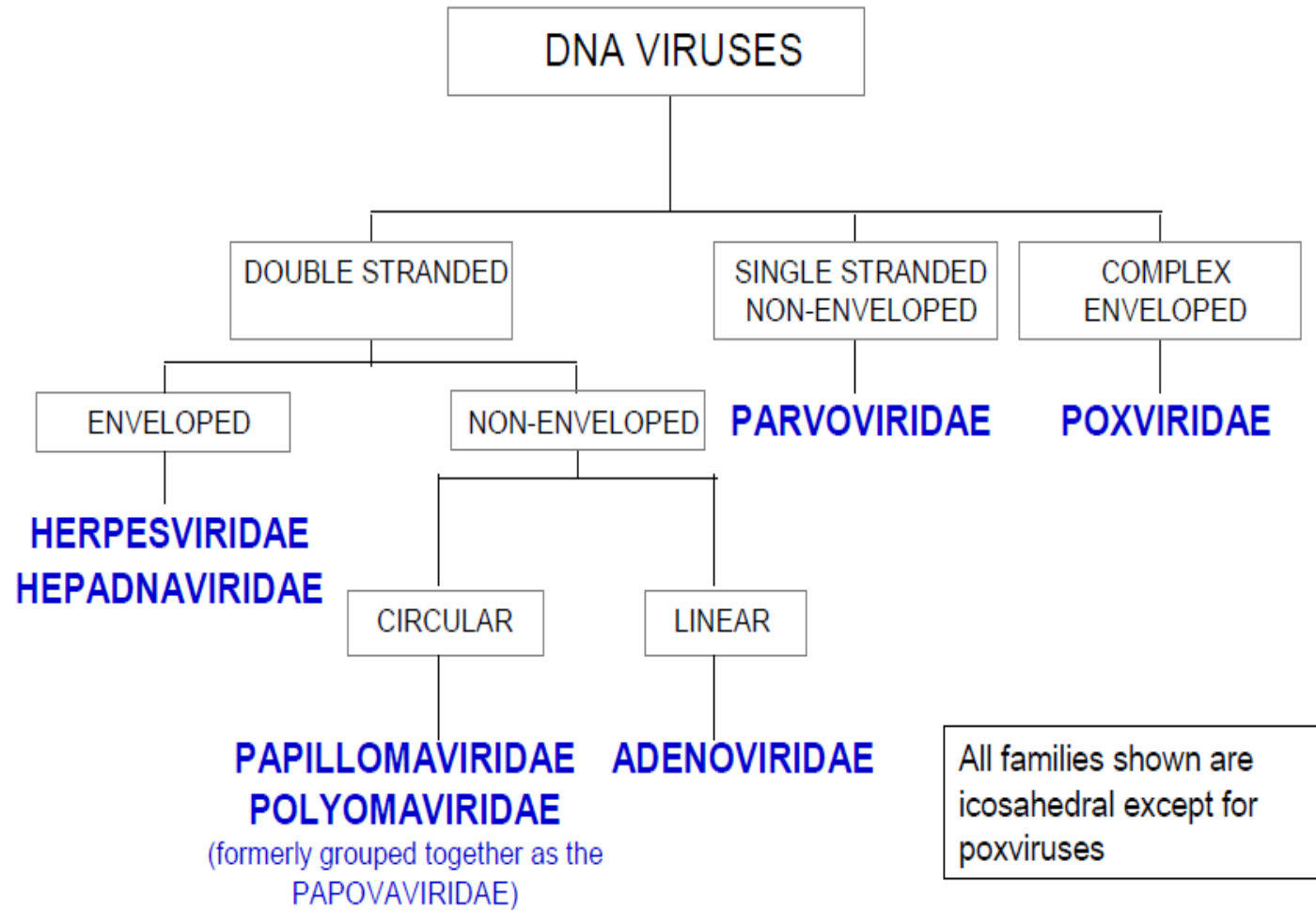
INTERNATIONAL CLASSIFICATION OF VIRUSES

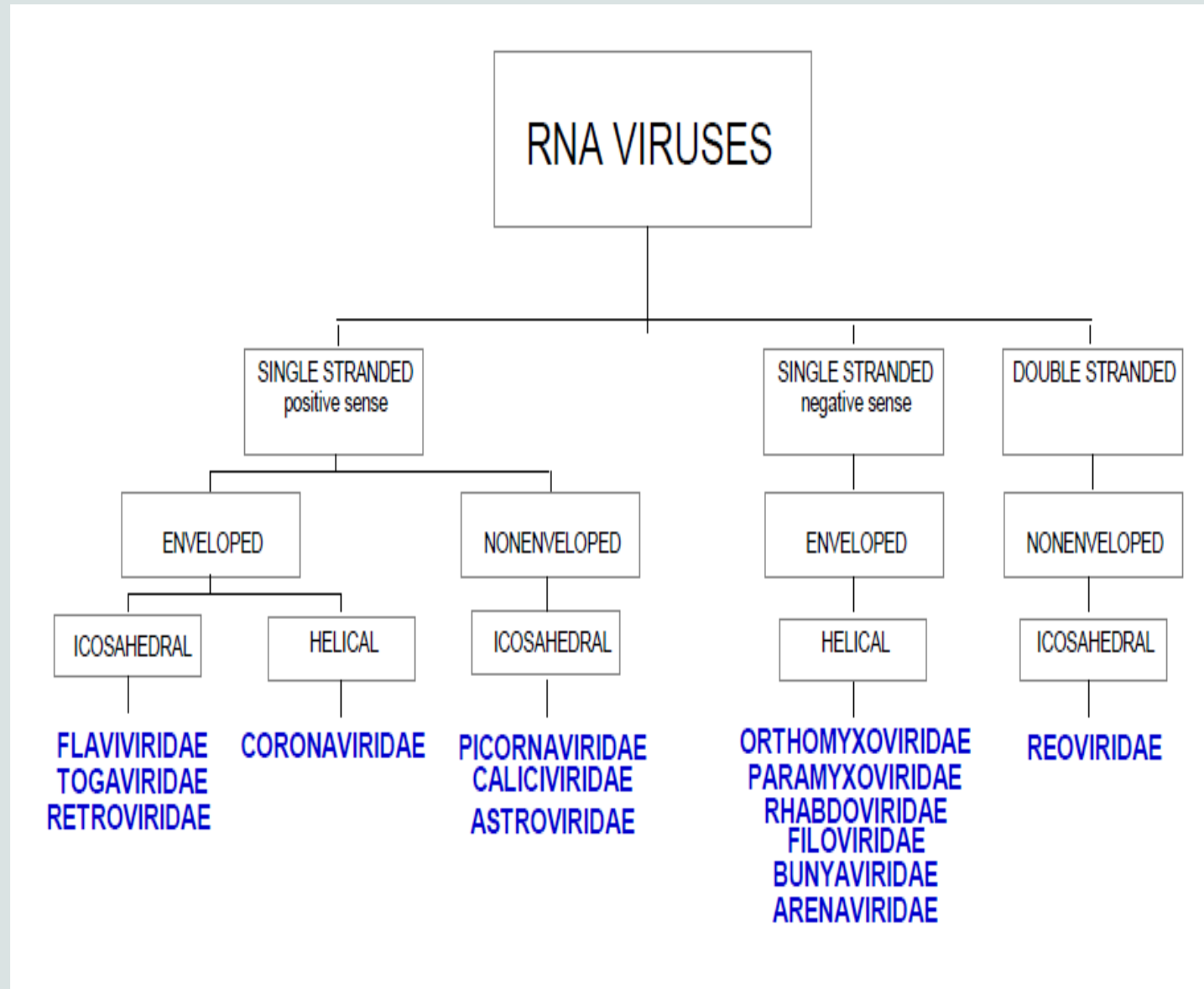
Nucleic acid

- RNA or DNA
- single-stranded or double-stranded
- Non segmented or segmented
- linear or circular if genome is single stranded RNA, can it function as mRNA?
- whether genome is diploid (it is in retroviruses)

Virion structure

- symmetry (icosahedral, helical, complex) number of capsomers if icosahedral
- enveloped or not





9. Virus Replication

As obligate intracellular parasites, Virus must enter and replicate in living cells in order to “reproduce” themselves. This “growth cycle” involves specific attachment of virus, penetration and uncoating, nucleic acid transcription, protein synthesis, maturation and assembly of the virions and their subsequent release from the cell by budding or lysis.

Steps of Viral Replication

➤ The following steps take place during viral replication

1. Recognition
2. Attachment
3. Penetration
4. Uncoating
5. Replication
6. Transcription
7. Protein Synthesis
8. Assembly
9. Lysis and release

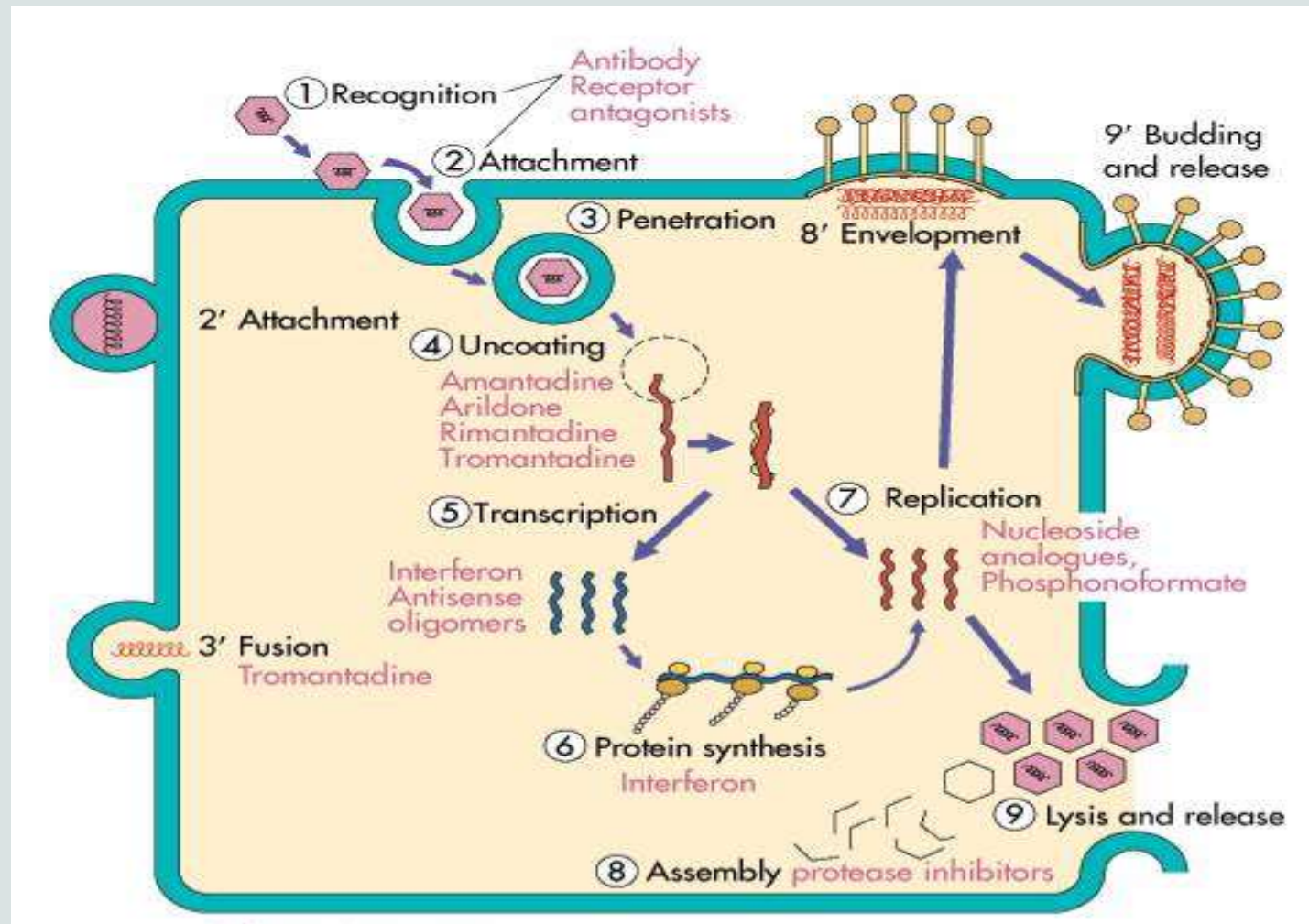


Figure 11: steps of virus replication

1. Attachment/Adsorption

- The virus uses specific proteins (ligands) on its surface to bind to receptors on the host cell. This interaction is highly specific and determines the virus's ability to infect certain cell types.
- Receptors can include glycoproteins, lipids, or other molecules that are typically involved in normal cellular functions, such as cell signaling or adhesion.
- The attachment process can also trigger conformational changes in the virus or the host cell, facilitating the next steps of entry.

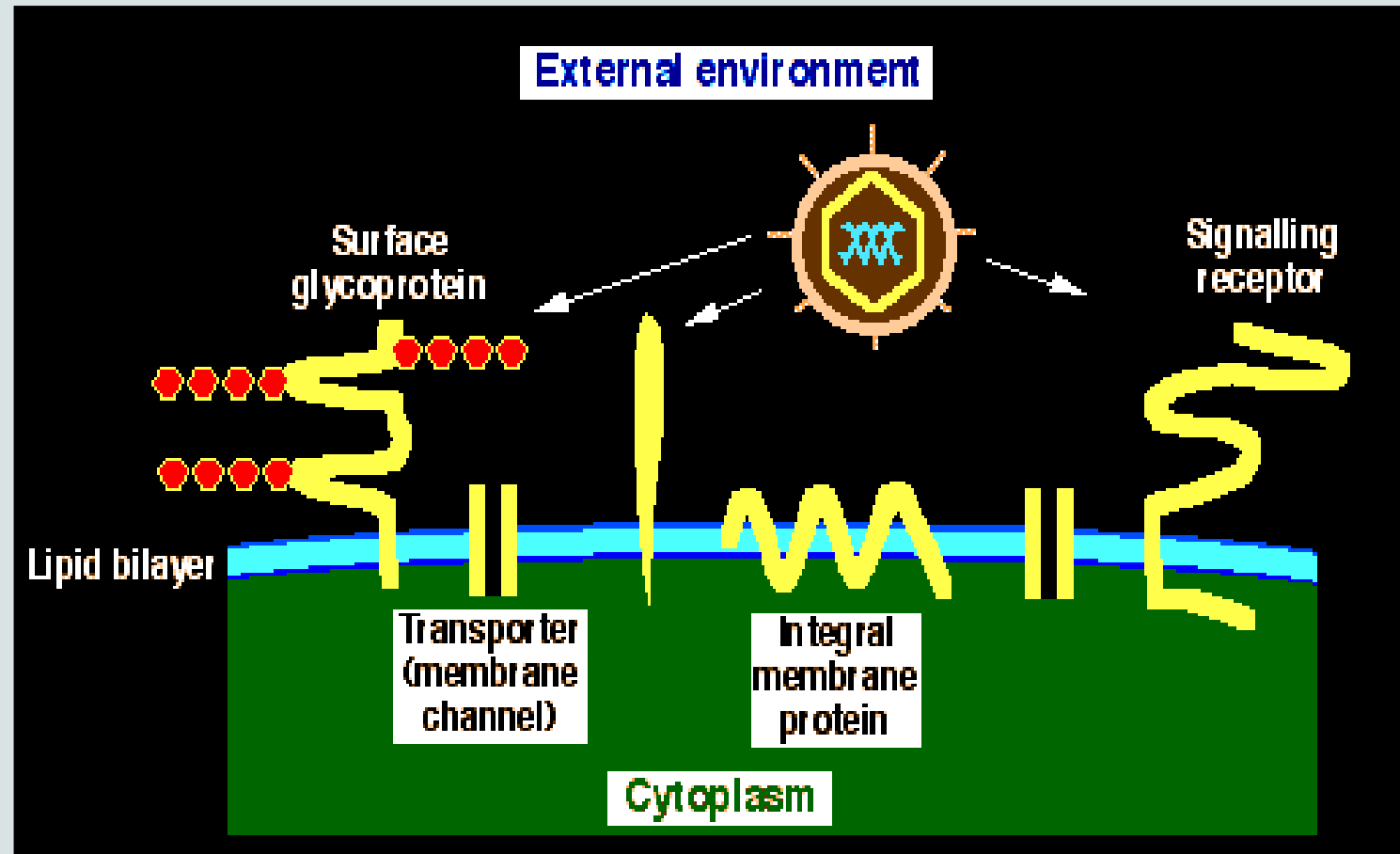
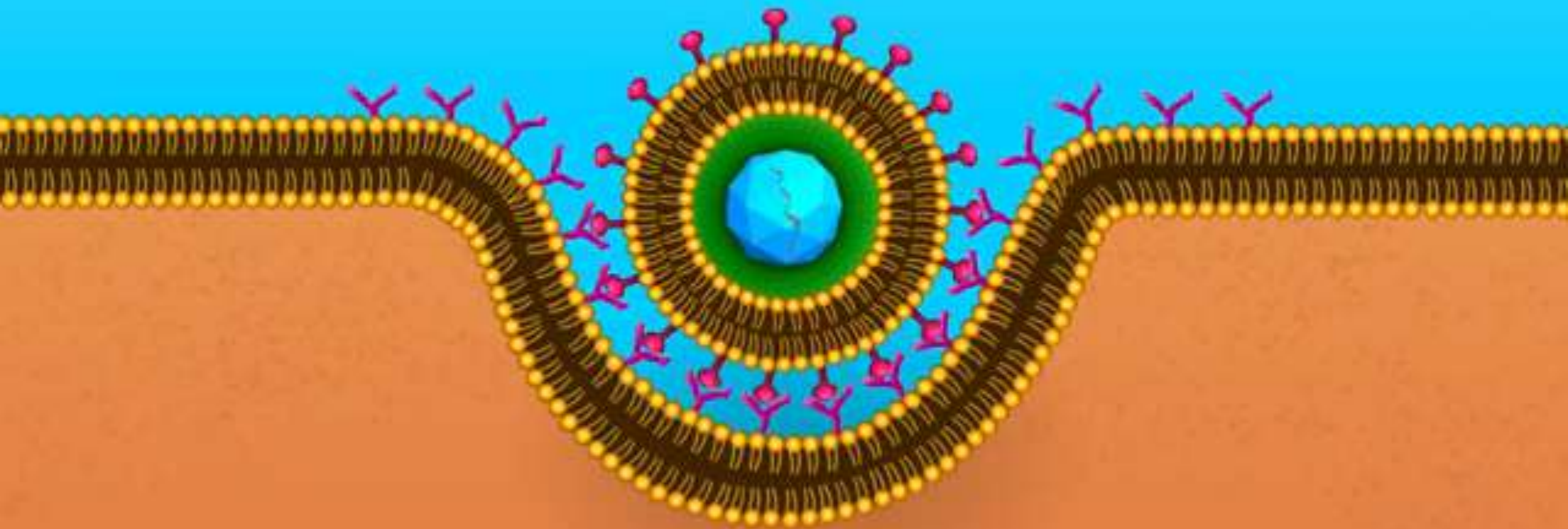


Figure 12: General overview of virus attachment



2. Penetration

- **Endocytosis:** The virus is engulfed by the host cell membrane, forming an endocytic vesicle. This process can be receptor-mediated, where the binding of the virus to its receptor initiates internalization.
- **Fusion:** Enveloped viruses can directly fuse with the host cell membrane. This often requires a specific pH environment or the presence of certain cellular proteins (fusion proteins) that facilitate membrane merging.
- Once inside, the endocytic vesicle may undergo acidification, which can trigger the release of the viral genome from the capsid.

3. Uncoating

- This step involves the disassembly of the viral capsid, releasing the viral genome into the cytoplasm or nucleus.
- Uncoating can occur through various mechanisms, including enzymatic degradation by host cell proteases or changes in pH within the endosome.
- The timing and location of uncoating are crucial, as they must coincide with the cell's replication machinery to effectively produce new viral components.

4- Replication

- For RNA viruses, replication often involves the synthesis of a complementary RNA strand using RNA-dependent RNA polymerase, which the virus may carry or encode.
- For DNA viruses, the viral DNA can be replicated using the host's DNA polymerases, often within the nucleus.
- Some viruses may employ different strategies, such as reverse transcription (e.g., retroviruses), where RNA is converted back to DNA before integration into the host genome.

5- Transcription and translation

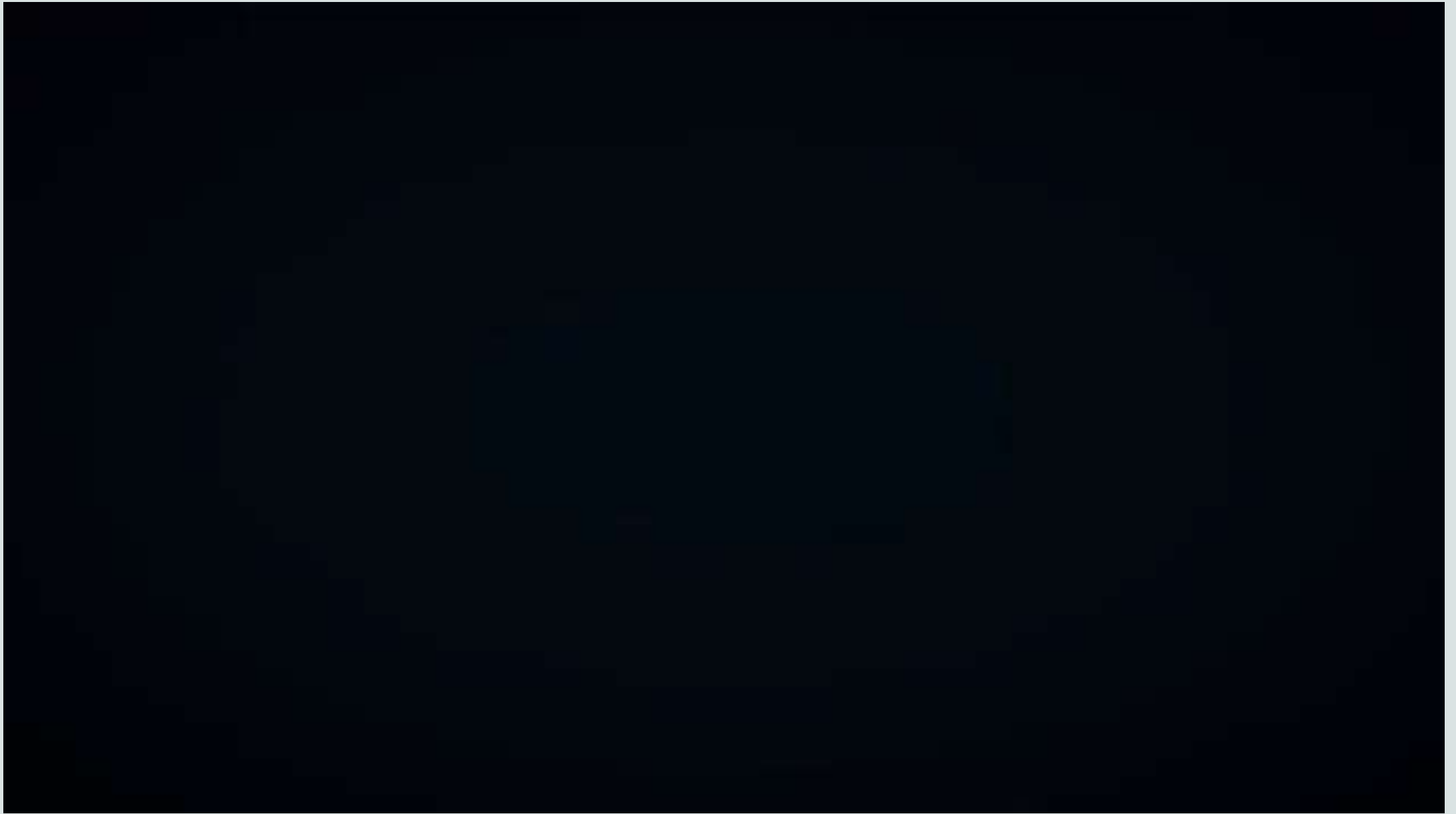
- The viral genome is transcribed into mRNA, which serves as a template for protein synthesis.
- Host ribosomes translate the viral mRNA into viral proteins. This process may involve post-translational modifications, such as glycosylation or phosphorylation, which are important for protein functionality.
- The types of proteins produced include structural proteins (for the viral capsid and envelope) and non-structural proteins (involved in replication, regulation, and evasion of the host immune response).

6. Assembly

- Newly synthesized viral genomes and proteins are assembled into new virions. This process typically occurs in specific cellular compartments, such as the cytoplasm or nucleus, depending on the virus type.
- Structural proteins encapsulated the viral genome, forming nucleocapsids, while envelope proteins may be inserted into the host cell membrane.
- The assembly process is often facilitated by viral proteins that direct the organization of components into a mature virion.

7. Release

- **Lysis:** Non-enveloped viruses often cause cell death by lysing the host cell, releasing the newly formed virions into the extracellular space to infect other cells.
- **Budding:** Enveloped viruses acquire their outer lipid membrane from the host cell membrane during release. This process allows the virus to exit the cell without immediately killing it, enabling continued viral replication within the host.
- The release of virions can also involve cellular pathways that help transport the newly formed particles to the cell surface.



10. Virus Pathogenicity

Virus pathogenicity refers to the ability of a virus to cause disease in a host organism. This involves several factors and mechanisms:

1. Virulence Factors: These are specific traits of the virus that enhance its ability to cause disease. This may include:

1. **Adhesion:** Ability to attach to host cells.

2. **Immune Evasion:** Mechanisms to evade or suppress the host's immune response (e.g., producing proteins that inhibit interferons).

2. Tissue Tropism: Different viruses have preferences for specific tissues or cell types, which influences the type of diseases they cause. For example, the influenza virus primarily infects respiratory tract cells.

3. Replication Rate: Fast-replicating viruses can overwhelm the host's immune system more quickly, leading to more severe disease.

4. Cytopathic Effects: Some viruses cause direct damage to host cells, leading to cell death and tissue damage. This can manifest as symptoms of the disease.

5. Immune Response: The host's immune response plays a crucial role in pathogenicity. A strong immune response can control the infection, while an inadequate or overactive response can lead to severe disease.

6. Transmission: The mode of transmission (e.g., respiratory droplets, bodily fluids) affects how widely a virus can spread and its overall impact on health.

7. Genetic Variability: High mutation rates in some viruses (like HIV and influenza) can lead to new strains that may evade the immune system, complicating treatment and prevention.

➤ Cycle of infection

A. Entry: Virus enters through mucous membranes or skin, Respiratory, oral, sexual, percutaneous (needles, wounds, bites).

B. Primary site replication

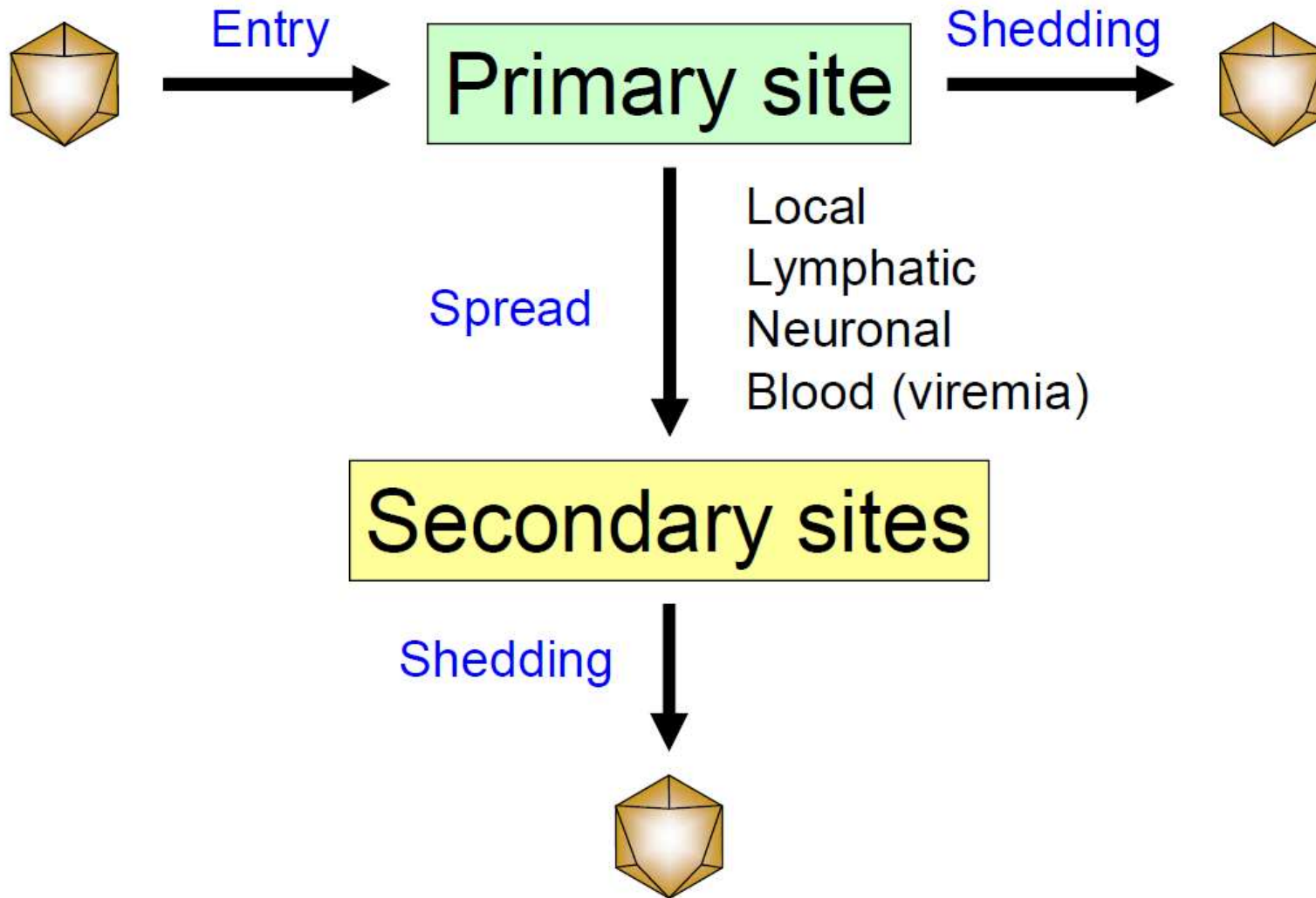
C. Spread: Local, blood (viremia), neuronal, and lymphatic.

D. Secondary site replication; target tissue

E. Shedding, transmission

- Routes: respiratory, gastrointestinal (oral-fecal), urogenital, skin
- Mechanisms
 - Indirect contact: aerosols, fomites
 - Direct contact; lesions, saliva, sex, animal or insect bites

Cycle of infection



11. Types of infection

➤ There are different types of infection:

1. Productive Infection:

- **Definition:** This occurs when a virus successfully enters a host cell and replicates, leading to the production of new infectious virus particles.
- **Outcome:** The host cell may eventually die due to the viral replication, but it also leads to the spread of the virus to other cells.

2. Abortive Infection:

- **Definition:** In this type, the virus enters a host cell but fails to replicate. This can occur due to various factors, such as the absence of necessary cellular machinery.
- **Outcome:** No new virus particles are produced, and the cell may remain intact or undergo apoptosis.

3. Latent Infection:

- **Definition:** The virus remains in a dormant state within the host cell after initial infection. It does not produce infectious virus immediately but can reactivate later.
- **Outcome:** The host may not show symptoms during the latent phase, but the virus can reactivate and lead to a productive infection at a later time.

4. Reactivation:

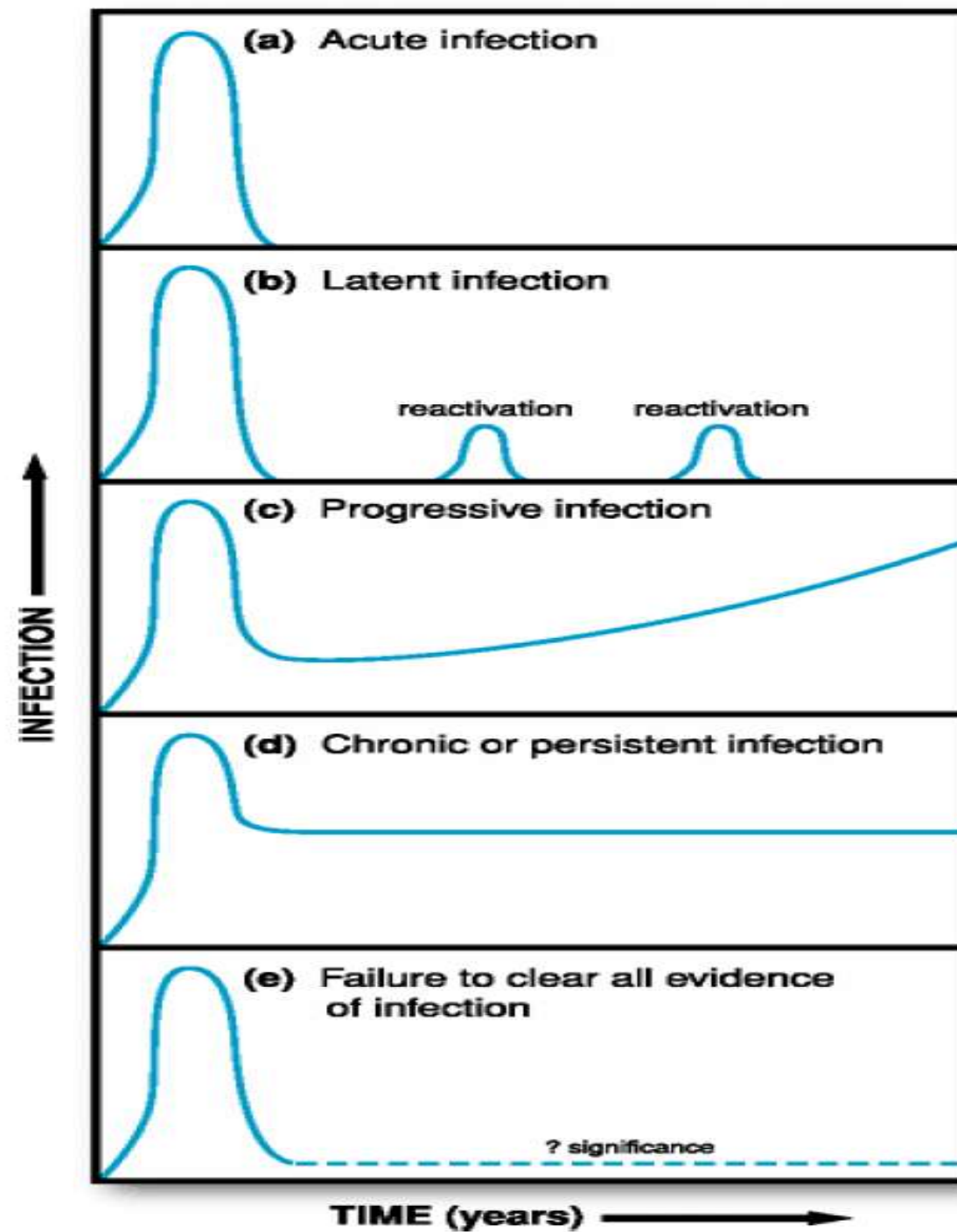
- **Definition:** This is the process by which a latent virus becomes active again, leading to the production of new virus particles.
- **Outcome:** Reactivation can be triggered by various factors, such as stress, immunosuppression, or other infections.

5. Chronic Infection:

- **Definition:** In chronic infections, the virus persists in the body for an extended period, often leading to continuous production of virus particles.
- **Outcome:** The host may experience ongoing symptoms or complications, and the infection can lead to diseases such as hepatitis or HIV/AIDS.

6. Acute Infection:

- **Definition:** This type of infection occurs suddenly and is typically characterized by a rapid onset of symptoms. The virus replicates quickly and may resolve in a short period.
- **Outcome:** The host may recover completely, develop immunity, or, in some cases, progress to a chronic infection.



12. Diseases

- Disease results from infection and can vary in severity. While many infections are harmless and do not lead to disease, understanding the molecular differences between infection and disease is a key focus in pathogenesis research. Highly virulent viruses like rabies, Ebola, and HIV are exceptions, causing significant disease in most infected individuals.
- Disease can manifest through various mechanisms, including:
 - **Cell and tissue destruction** (e.g., rabies virus killing neurons).
 - **Inflammatory cytokine induction** (e.g., fever from many viruses).
 - **Cellular dysfunction** (e.g., LCMV affecting the pituitary).
 - **Tumor induction.**
 - **Immune response effects** (e.g., immunopathology).
- Common symptoms of viral disease, such as fever and malaise, are nonspecific and provide limited insight into pathogenesis. In contrast, specific symptoms (e.g., hepatitis, paralysis) offer valuable clues about the underlying pathogenic processes and link clinical observations to the science of viral pathogenesis.

13. Virulence

- Virulence refers to a virus's ability to cause disease and influences the infection-disease relationship. More virulent viruses lead to disease in a higher proportion of infected individuals and typically cause more severe symptoms. Virulence can manifest in various ways, including:
 - Rapid death** (e.g., variola major causing smallpox).
 - Tumor induction** over extended periods (e.g., certain papillomaviruses and herpesviruses).
 - Organ failure** over many years (e.g., chronic HBV or HCV infections).
- The specific manifestations of virulence depend on the unique strategies employed by each virus during infection.

14. Viral Vaccines

A. Live, attenuated

- ✓ Naturally or artificially reduced in pathogenicity in humans

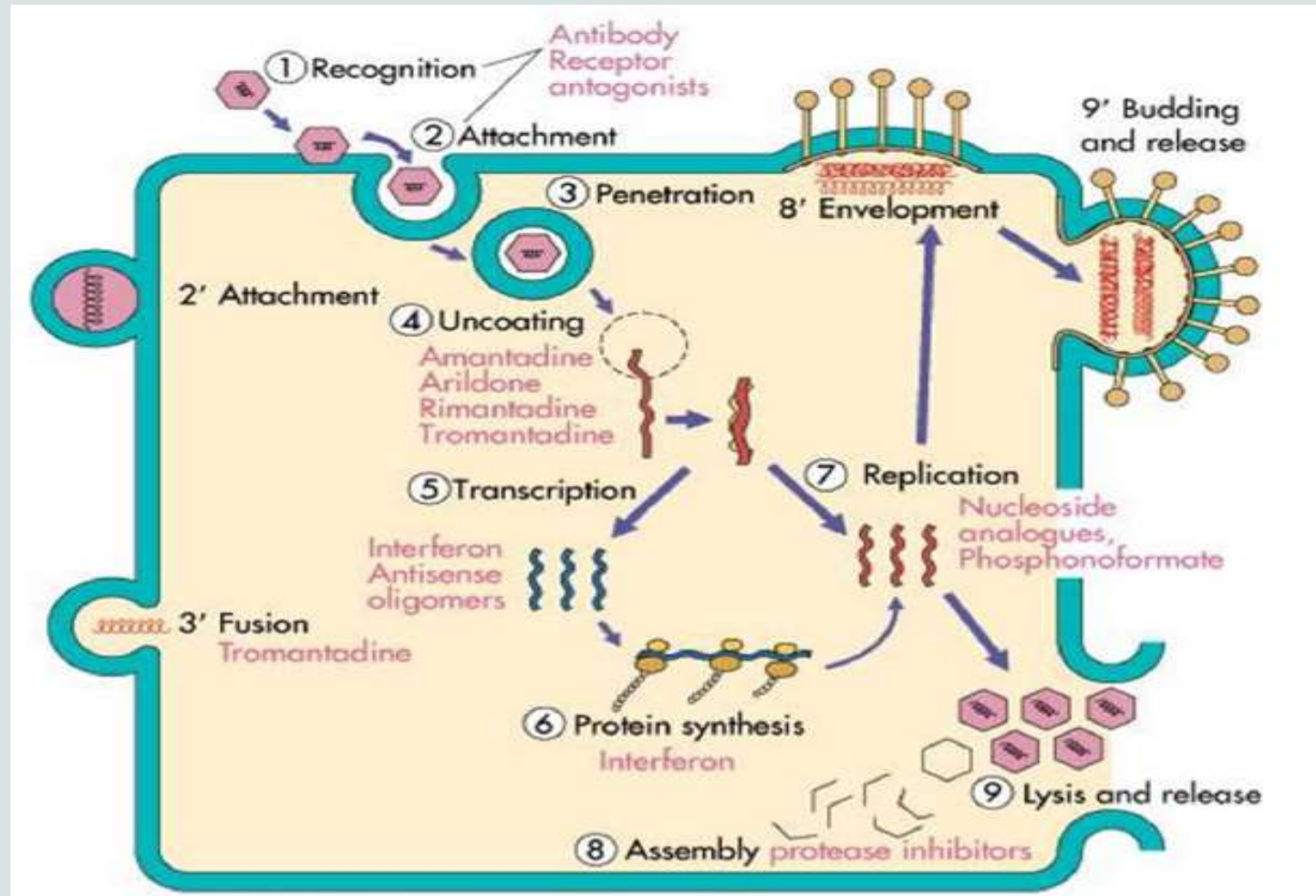
B. Killed, inactivated

- ✓ Wild type virus, chemically inactivated

C. Subunit

- ✓ Part of the virus, either natural or synthetic

15. Antiviral drugs



- Antiviral drugs work through various mechanisms to inhibit viral replication and reduce the severity of infections. Here are the primary mechanisms by which antiviral drugs function:

1.Inhibition of Viral Entry:

- Some antiviral drugs prevent viruses from entering host cells by blocking viral attachment to cell receptors or interfering with the fusion of the viral envelope with the host cell membrane.

2.Inhibition of Viral Uncoating:

- Certain antivirals disrupt the uncoating process, which is necessary for the release of viral genetic material into the host cell. By preventing uncoating, these drugs inhibit the virus's ability to replicate.

3.Nucleoside and Nucleotide Analogues:

- These drugs mimic the building blocks of viral DNA or RNA, leading to the incorporation of faulty nucleotides during viral replication. This results in premature termination of the viral genome or errors that inhibit viral function (e.g., acyclovir, zidovudine).

4. Inhibition of Viral Polymerases:

- Some antivirals specifically inhibit viral polymerases (enzymes responsible for viral genome replication), effectively blocking the synthesis of viral nucleic acids. Examples include sofosbuvir for hepatitis C and foscarnet for herpesviruses.

5. Inhibition of Viral Proteases:

- Protease inhibitors prevent the cleavage of viral polyproteins into functional proteins, disrupting the maturation and assembly of new virions. This mechanism is commonly used in HIV and hepatitis C treatments (e.g., ritonavir, boceprevir).

6. Inhibition of Viral Assembly and Release:

- Certain antivirals interfere with the assembly of viral components or prevent the release of new virions from infected cells, thereby limiting the spread of the virus (e.g., neuraminidase inhibitors like oseltamivir for influenza).

7. Modulation of the Immune Response:

- Some antiviral agents enhance the host's immune response against the virus, promoting the clearance of the virus from the body. Interferons are examples of such agents used in hepatitis and some viral infections.

8. Targeting Viral Enzymes:

- Antiviral drugs may target specific viral enzymes that are essential for the virus's life cycle, effectively blocking critical processes needed for replication and infection.

16. Viral Hepatitis

➤ **Definition and Characteristics:**

- Hepatitis is an inflammation of the liver, recognized for thousands of years.
- Clinically characterized by:
 - Jaundice
 - Abdominal pain
 - Liver enlargement
 - Sometimes fever

➤ **Causes:**

- Caused by viral and non-viral agents, including alcohol.
- Viral hepatitis is a systemic infection primarily affecting the liver.

➤ **Types of Hepatitis Viruses:**

1.Hepatitis A Virus (HAV): Infectious hepatitis.

2.Hepatitis B Virus (HBV): DNA virus.

3.Hepatitis C Virus (HCV): Non-A, non-B hepatitis.

4.Hepatitis D Virus (HDV): Delta hepatitis.

5.Hepatitis E Virus (HEV): Enterically transmitted non-A, non-B.



Thank you

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